Ficha 4 (Resposta em Frequência - diagramas de Bode)

1.

$$u_1 = 2 \operatorname{sen}(\underbrace{0{,}2}_{\omega} t)$$

$$y(t) = B \cdot A \operatorname{sen}(\omega t - \phi) = 10 \times 2 \operatorname{sen}(0{,}2t - 0^\circ)$$

$$B = 20\,dB \qquad = 20 \operatorname{sen}(0{,}2t)$$

$$20 = 20 \log B$$

$$\Leftrightarrow B = 10$$

$$\phi = -\arctan\left(\frac{\omega}{\omega_b}\right) = 0^\circ$$

2.

$u_2(t) = 10\,\text{sen}(20t)$

$y(t) = B.A\,\text{sen}(wt - \phi) = 45\,\text{sen}(20t - 63^\circ)$

$B = 13\,dB \qquad 13 = 20 \log B$

$\phi = -63^\circ \qquad \Leftrightarrow B = 10^{13/20} = 4{,}5$

B.

$Q_3(t) = 5\,\text{sen}(300t + 10°)$

$y(t) = B.A\,\text{sen}(\omega t + \phi) = 0{,}31 \times 5\,\text{sen}(300t + 10 - 88) = 1{,}6\,\text{sen}(300t - 78°)$ //

$B = -10\,dB$

$\phi = -88°$

$-10 = 20 \log B$

$\Leftrightarrow B = 10^{-10/20} = 0{,}31$

4.

$u_4(t) = u_1(t) + u_2(t) + u_3(t)$

$y(t) = y_1(t) + y_2(t) + y_4(t)$

$\Leftrightarrow y(t) = 20 \sin(0{,}2t) + 45 \sin(20t - 63^\circ) + 1{,}6 \sin(300t - 78^\circ)$ //

5.

5.1.

$G(s) = \frac{Y(s)}{X(s)}$

$\Leftrightarrow G(s) = \frac{K_H}{Ts+1}$

$G(j\omega) = \frac{K_H}{j\frac{\omega}{\omega_0}+1}$

$|G(j\omega)| = \frac{K_H}{\sqrt{(\frac{\omega}{\omega_0})^2+1}}$ //

ganho em dB = $20 \log |G(j\omega)|$

$\omega < \omega_0$
ganho = 20dB
$\frac{\omega}{\omega_0} \approx 0$

$\Leftrightarrow 20 = 20 \log\left(\frac{K_H}{\sqrt{(\frac{\omega}{\omega_0})^2+1}}\right)$

$\Leftrightarrow 1 = \log\left(\frac{K_H}{\sqrt{(\frac{\omega}{\omega_0})^2+1}}\right)$

$\Leftrightarrow 10 = \frac{K_H}{\sqrt{0+1}}$

$\Leftrightarrow K_H = 10$ //

5.2.

$\frac{1}{\tau} = 10$

$\Leftrightarrow \tau = \frac{1}{10}$

$\Leftrightarrow \tau = 0{,}1\ s$ //

6.

$u(t) = 2u_0(t)$

$Y(s) = G(s)X(s)$

$\Leftrightarrow Y(s) = \frac{K_{rp}}{\tau s+1} \times \frac{2}{s}$

$\Leftrightarrow Y(s) = \frac{2K_{rp}}{s(\tau s+1)}$

$\Leftrightarrow Y(s) = \frac{2\times 10}{s(0{,}1s+1)}$

$\Leftrightarrow Y(s) = \frac{20}{s(0{,}1s+1)}$

$\Leftrightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{20}{s(0{,}1s+1)}\right\}$

$\Leftrightarrow y(t) = 20\,\mathcal{L}^{-1}\left\{\frac{1}{s(0{,}1s+1)}\right\}$

$\Leftrightarrow y(t) = 20\,\mathcal{L}^{-1}\left\{\frac{1}{0{,}1s(s+\frac{1}{0{,}1})}\right\}$

$\Leftrightarrow y(t) = 200\,\mathcal{L}^{-1}\left\{\frac{1}{s(s+\frac{1}{0{,}1})}\right\}$

$\Leftrightarrow y(t) = \cancel{200}\times\frac{1}{\cancel{10}}(1-e^{-10t})$

$\Leftrightarrow y(t) = 20(1-e^{-10t})$ //